1 Preliminary notions

1.1 Probability theory

The foundation of modern probability theory is based on measure theory.

Definition 1.1 (σ-algebra)**.** A **σ-algebra** $\mathcal{F}$ on a set Ω is a collection of subsets of Ω that

- $\Omega \in \mathcal{F}$
- if $A \in \mathcal{F}$ then $\Omega \setminus A \in \mathcal{F}$
- if $A_1, A_2, \ldots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

Definition 1.2 (Measure)**.** Let Ω be a set and $\mathcal{F}$ a σ-algebra on Ω. A function $\mu : \mathcal{F} \to [0, \infty]$ is **measure** if satisfies

1. $\mu(\emptyset) = 0$
2. $\mu(A) \geq 0$ for all $A \in \mathcal{F}$
3. if $\{A_i\}_{i=1}^{\infty} \in \mathcal{F}$ and $A_i \cap A_j = \emptyset$ for all $i \neq j$ then
$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu\left(A_i\right)$$

If μ is a measure on $\mathcal{F}$ such that $\mu(\Omega) = 1$, then μ is called a **probability measure**.

Definition 1.3 (Probability space)**.** A probability space is a triple $(\Omega, \mathcal{F}, P)$ where

- Ω is the set of all possible outcomes
- $\mathcal{F}$ is a σ-algebra on Ω
- P is a probability measure.

For a more details in probability theory and stochastic processes, see [4] or [5].

1.2 Stochastic processes

Definition 1.4 (Stochastic processes)**.** A stochastic process is an indexed family of random variables
$$\{X(t, \omega)\}_{t \in T},$$
on a probability space $(\Omega, \mathcal{F}, P)$ assuming values in $\mathbb{R}^n$.

If the set T is countable then the stochastic process is said to be *discrete*, if T is some interval of $\mathbb{R}$ then it is *continuous*.

An intuitive way to understand stochastic processes is to consider what is $X(t, \omega)$ when t or ω is fixed. If we fix $t \in T$ then the map
$$\omega \to X_t(\omega),$$
for $\omega \in \Omega$ is a random variable. If $\omega \in \Omega$ is fixed then the map
$$t \to X_\omega(t),$$
for $t \in T$ is a deterministic function, and is called *path* of $X(t, \omega)$.

Definition 1.5 (Filtration)**.** A **filtration** is a family of σ-algebras $\{\mathcal{F}_t\}_{t\in T}$ indexed by T, such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $s \leq t$, where $s, t \in T$.

The idea is that a filtration $\mathcal{F}_t$ represents the information available from the past of an stochastic process up to time t. Thus we define a **filtered probability space** as a probability space with a filtration $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in T}, P)$.

Definition 1.6 (Adapted process)**.** Let X_t be a stochastic process on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in T}, P)$. The process is **adapted** to the filtration $\{\mathcal{F}_t\}_{t\in T}$ if, for every $t \in T$, the random variable X_t is $\mathcal{F}_t$-measurable.

The idea of an adapted process is key when modeling using stochastic processes, as it ensures that the process at time t depends only on the information available up to that time, but not on future information.

1.2.1 Martingales

A martingale is a fundamental concept in the theory of stochastic processes, formalizes the idea of a "fair game", where given all the information up to now, the best prediction for the future is simply the present value.

Definition 1.7 (Martingale)**.** A stochastic process X_t where $t \in T$ is called a **martingale** with respect to a filtration $\{\mathcal{F}_t\}_{t\in T}$ if it satisfies the following conditions:

1. $E[|X_t|] < \infty$, for each $t \in T$
2. X_t is measurable with respect to $\mathcal{F}_t$ for each $t \in T$
3. If X_t is discrete
$$\mathbb{E}[X_{t+1}|\mathcal{F}_t] = X_t \quad \forall t \in \mathbb{N},$$
if X_t is continuous
$$\mathbb{E}[X_t|\mathcal{F}_s] = X_s \quad \forall s \leq t.$$

1.2.2 Brownian Motion

Brownian motion (also called Wiener process) is the fundamental stochastic process underlying continuous-time financial modeling. Formally, a Brownian motion $\{W_t\}_{(t\geq 0)}$ is defined by the following properties:

1. $W_0 = 0$
2. The map $t \to W_t(\omega)$ is almost surely continuous
3. W_t has independent increments, i.e. for $t_1 < t_2 < t_3 < t_4$, $W_{t_2} - W_{t_1}$ and $W_{t_4} - W_{t_3}$ are independent random variables
4. W_t has gaussian increments, i.e. $W_t - W_s \sim \mathcal{N}(0, t-s)$ for $0 \leq s \leq t$

where $\mathcal{N}(\mu, \sigma^2)$ denotes the normal distribution with mean μ and variance σ^2.

The Brownian motion has some key features that will need afterwards, for example it can be shown that the paths $t \to W_t(\omega)$ although are almost surely continuous, they are nowhere differentiable.

Also it can be shown that the Wiener process is an almost surely continuous martingale with quadratic variation $[W,W]_t = t$. These two properties together with $W_0 = 0$ form an equivalent definition of the Wiener process, known as Lévy's characterization.

Proofs of these properties, the equivalence between the two definitions, and the existence of Brownian motion can be found in [7] and [8, Ch. 2].

2 Stochastic calculus

To model the dynamics of a financial asset's value, an ordinary differential equation of the form

$$\frac{dS_t}{dt} = f(t, S_t)$$

is insufficient: it cannot capture the random fluctuations inherent in financial data. Stochastic differential equations (SDEs) add a noise term to handle this,

$$dS_t = f(t, S_t)\,dt + \sigma(t, S_t)\,dW_t, \tag{1}$$

where W_t is a Brownian motion. In integral form this reads

$$S_t = S_0 + \int_0^t f(s, S_s)\,ds + \int_0^t \sigma(s, S_s)\,dW_s, \tag{2}$$

thus a key step is to give rigorous meaning to the stochastic integral $\int \sigma\,dW_s$.

The most well known process that has a well defined stochastic integral is the Wiener process or Brownian motion, this stochastic integral is known as the Ito integral and will allow us to have the machinery to study SDEs and thus model the dynamics of financial assets.

2.1 Ito's integral

Definition 2.1. Let $\mathcal{V} = \mathcal{V}(S, T)$ the class of functions

$$f(t, \omega) : [0, \infty) \times \Omega \to \mathbb{R},$$

such that

- (t, ω) is $\mathcal{B} \times \mathcal{F}$-measurable, where $\mathcal{B}$ is the Borel σ-algebra on $[0, \infty)$ and $\mathcal{F}$ is the σ-algebra on Ω,
- $f(t, \omega)$ is $\mathcal{F}_t$ adapted.
- $\mathbb{E}\left[\int_S^T f(t, \omega)^2 dt\right] < \infty$.

We will start by defining the Itô integral for *elementary functions*, then this can be extended to functions in $\mathcal{V}$.

Definition 2.2. A function ϕ is called *elementary* if it has the form

$$\phi(t, \omega) = \sum_j e_j(\omega)\mathbb{1}_{[t_j, t_{j+1})}(t).$$

For elementary functions we define the Ito integral as

$$\int_S^T \phi(t, \omega)\,dB_t(\omega) = \sum_j e_j(\omega)\left[B_{t_{j+1}} - B_{t_j}\right](\omega).$$

Lemma 2.1 (Ito isometry)**.** Let $\phi(t, \omega)$ be a bounded elementary function, then

$$\mathbb{E}\left[\left(\int_S^T \phi(t, \omega) dB_t(\omega)\right)^2\right] = \mathbb{E}\left[\int_S^T \phi(t, \omega)^2 dt\right].$$

Using the Ito isometry and the definition of the Ito integral for elementary functions, the Ito integral can be extended to the class of functions $\mathcal{V}$, using that the functions in $\mathcal{V}$ can be expressed as the limit of sequences of simpler functions. It is worth mentioning that the Itô isometry still holds for any function in $\mathcal{V}$.

The result of the integral is a stochastic process, which is a continuous martingale. This is similar to regular integration, where the result of integrating a function is a new function, in this case the result of integrating a process "over" a stochastic process is a new stochastic process, plus the resulting process has the nice property of being a martingale.

The class $\mathcal{V}$ is rather limited and does not even contain all continuous functions, see [9]. Having to restrict to this class would be rather inconvenient when modeling using SDEs, luckily the Itô integral can be extended to a broader class by relaxing the last condition of Definition 2.1

$$P\left(\int_S^T f(t,\omega)^2 dt < \infty\right) = 1,$$

will call this class of functions $\mathcal{W}$. Notice that the class $\mathcal{W}$ does contain all continuous functions.

2.2 Ito's formula

Having defined the Ito integral, would be interesting to have a way to easily compute stochastic integrals without having to compute the limit of elementary functions, that is the purpose of the Ito's lemma or Ito's formula. The Ito's formula is a stochastic version of the chain rule for derivatives.

Definition 2.3 (Ito process). Let B_t be a Brownian motion on $(\Omega, \mathcal{F}, P)$. An Ito process is a stochastic process X_t on $(\Omega, \mathcal{F}, P)$ of the form

$$X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s, \tag{3}$$

where μ and σ are adapted, measurable processes, that satisfy the integrability conditions

$$P\left(\int_0^T |\mu(t,\omega)|\, dt < \infty\right) = 1,$$

and

$$P\left(\int_0^T \sigma(t,\omega)^2 dt < \infty\right) = 1.$$

The equation (3) can be also written in differential form as

$$dX_t = \mu\, dt + \sigma\, dB_t.$$

Theorem 2.2 (1-dimensional Itô's formula). Let X_t be an Itô process given by

$$dX_t = \mu\, dt + \sigma\, dB_t.$$

Let $g(t,x) \in C^2([0,\infty) \times \mathbb{R})$, then

$$Y_t = g(t, X_t),$$

is an Ito process given by

$$dY_t = \frac{\partial g}{\partial t}(t, X_t) dt + \frac{\partial g}{\partial x}(t, X_t) dX_t + \frac{1}{2}\frac{\partial^2 g}{\partial x^2}(t, X_t)(dX_t)^2, \tag{4}$$

where $(dX_t)^2$ is computed according to the rules

$$(dB_t)^2 = dt, \qquad (dt)^2 = 0, \qquad dB_t \cdot dt = 0.$$

For the proof of the Ito's formula we refer to [8]. The formula can be extended to multiple dimensions.

Definition 2.4 (n-dimensional Ito process)**.** Let $B_t = (B_1(t), \dots, B_m(t))$ be a m-dimensional Brownian motion, let μ_i and σ_{ij} be adapted, measurable processes such that

$$P\left(\int_0^T |\mu_i(t,\omega)|\, dt < \infty\right) = 1, \qquad P\left(\int_0^T \sigma_{ij}(t,\omega)^2 dt < \infty\right) = 1,$$

then we define the n-dimensional Ito process $X_t = (X_1(t), \dots, X_n(t))$ as

$$\begin{cases} dX_1(t) = \mu_1 dt + \sigma_{11} dB_1(t) + \cdots + \sigma_{1m} dB_m(t), \\ \vdots \\ dX_n(t) = \mu_n dt + \sigma_{n1} dB_1(t) + \cdots + \sigma_{nm} dB_m(t), \end{cases}$$

or in matrix notation

$$dX_t = \mu\, dt + \sigma\, dB_t,$$

where $\mu = (\mu_1, \dots, \mu_n)$ and $\sigma = (\sigma_{ij})$ are an n-dimensional vector and an $n \times m$ matrix respectively.

Theorem 2.3 (General Itô's formula)**.** Let X_t be an n-dimensional Itô process given by

$$dX_t = \mu\, dt + \sigma\, dB_t.$$

Let $g(t,x) = (g_1(t,x), \dots, g_p(t,x)) \in C^2([0,\infty) \times \mathbb{R}^n)$, then

$$Y_t = g(t, X_t),$$

is a p-dimensional Ito process whose components are given by

$$dY_k(t) = \frac{\partial g_k}{\partial t}(t, X_t)dt + \sum_i \frac{\partial g_k}{\partial x_i}(t, X_t) dX_i(t) + \frac{1}{2} \sum_{ij} \frac{\partial^2 g_k}{\partial x_i \partial x_j}(t, X_t) dX_i(t) dX_j(t), \quad (5)$$

where $dX_i(t)dX_j(t)$ is computed according to the rules

$$(dB_i(t))^2 = dt, \qquad dB_i(t)\, dB_j(t) = 0 \quad (i \neq j).$$

2.3 Stochastic differential equations

A **stochastic differential equation** (SDE) is an equation of the form

$$dX_t = b(t, X_t)\, dt + \sigma(t, X_t)\, dB_t, \qquad X_0 = x_0, \quad (6)$$

where $b : [0,T] \times \mathbb{R}^n \to \mathbb{R}^n$ is the *drift* and $\sigma : [0,T] \times \mathbb{R}^n \to \mathbb{R}^{n \times m}$ is the *diffusion* coefficient. A solution is an Itô process X_t satisfying (6) almost surely.

The following theorem, whose proof may be found in [8, Thm. 5.2.1], gives sufficient conditions for a solution to exist and be unique.

Theorem 2.4 (Existence and uniqueness)**.** Let $T > 0$. Suppose b and σ satisfy, for all $t \in [0, T]$ and $x, y \in \mathbb{R}^n$,

1. (Lipschitz) $|b(t, x) - b(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq C\,|x - y|$,
2. (Linear growth) $|b(t, x)|^2 + |\sigma(t, x)|^2 \leq C^2(1 + |x|^2)$,

for some constant $C > 0$, and let $x_0 \in \mathbb{R}^n$. Then the SDE (6) has a unique (up to indistinguishability) continuous adapted solution $\{X_t\}_{0 \leq t \leq T}$.

We close with two fundamental examples, both following [8, Ch. 5].

2.3.1 Example: Geometric Brownian motion

Let $\mu, \sigma \in \mathbb{R}$ with $\sigma \neq 0$ and $S_0 > 0$. Consider the SDE

$$dS_t = \mu S_t\,dt + \sigma S_t\,dB_t. \tag{7}$$

This is called **geometric Brownian motion** and is the model used in Section 3.2 for the dynamics of a stock price.

Applying Theorem 2.2 to $g(t, x) = \ln x$, with $g_t = 0$, $g_x = x^{-1}$, $g_{xx} = -x^{-2}$,

$$d(\ln S_t) = \frac{1}{S_t}\,dS_t - \frac{1}{2S_t^2}(dS_t)^2 = \mu\,dt + \sigma\,dB_t - \frac{\sigma^2}{2}\,dt = \left(\mu - \frac{\sigma^2}{2}\right)dt + \sigma\,dB_t.$$

Integrating both sides from 0 to t gives $\ln(S_t/S_0) = (\mu - \sigma^2/2)\,t + \sigma B_t$, so the unique solution is

$$S_t = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t\right]. \tag{8}$$

2.3.2 Example: Ornstein-Uhlenbeck process

Let $\theta, \sigma > 0$ and $\mu \in \mathbb{R}$. Consider the SDE

$$dX_t = \theta(\mu - X_t)\,dt + \sigma\,dB_t, \qquad X_0 = x_0. \tag{9}$$

This is the **Ornstein-Uhlenbeck (OU) process**, a mean-reverting model: the drift $\theta(\mu - X_t)$ pulls the process toward the long-run mean μ at rate θ. It appears, for instance, in the Vasicek interest rate model.

To solve, let $Y_t = (X_t - \mu)e^{\theta t}$. Applying Theorem 2.2 with $g(t, x) = (x - \mu)e^{\theta t}$,

$$dY_t = \theta(X_t - \mu)e^{\theta t}\,dt + e^{\theta t}\,dX_t = e^{\theta t}\big[\theta(\mu - X_t) + \theta(X_t - \mu)\big]dt + \sigma e^{\theta t}\,dB_t = \sigma e^{\theta t}\,dB_t.$$

Integrating and reverting the substitution,

$$X_t = \mu + (x_0 - \mu)\,e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)}\,dB_s. \tag{10}$$

The OU process is Gaussian with $\mathbb{E}[X_t] = \mu + (x_0 - \mu)e^{-\theta t} \to \mu$ as $t \to \infty$.

3 Black-Scholes

Black-Scholes is a mathematical model for pricing options. Although many other models have been developed since, the Black-Scholes model remains as the most important model in option pricing, as in many cases options in markets are quoted using the Black-Scholes formula.

A derivative is a financial instrument whose payments are derived from the price of an other asset, called the underlying asset. Derivatives are widely used in finance, whether to hedge financial risk, this is to protect against adverse price movements, or to bet on specific views, especially when it's difficult to take a position in the underlying asset. So being able to price them and justify the price is key for the market to function.

3.1 Replication and arbitrage

One of the most important concepts in modern finance is the so called *law of one price*. The idea is that if two financial assets have exactly the same payoffs in all possible scenarios, then they should have the same price.

This is a linked to the concept of *arbitrage*, informally we say that arbitrage is the opportunity to make a riskless profit. For example, if two assets that have exactly the same payoffs in all possible scenarios, but don't follow the law of one price, one can buy the cheaper one and sell the more expensive one, gaining and instant profit and compensating the payoffs of the sold asset using the payoffs of bought one, which match exactly.

We will work under the assumption that we are in operating in a market with no arbitrage, this assumption is technically known as *The First Fundamental Theorem of Asset Pricing*, and a key part of the *Efficient Market Hypothesis*. Under this assumption, we can price any derivative by creating a replicating portfolio, this is a set of assets that has exactly the same payoffs as the derivative in all possible scenarios.

There are two main ways of creating a replicating portfolio:

- **Static replication** refers to an replicating portfolio that is created at the beginning and held as is until maturity.
- **Dynamic replication** refers to a replicating portfolio that requires a continuous change of its composition over time.

For a further discussion on replication, and its applications in finance, we recommend the book [3].

We will show two examples of static replication from [9] that will be useful to understand the Black-Scholes model.

3.1.1 Example: Forward contract

Assume that there is an agent that borrows and lends money at a continuously compounded rate r, known as the **risk-free rate**. This means that if we borrow 1\$ at time $t = 0$, we will have to pay e^{rT}\$ at time $t = T$.

A forward contract gives the holder the obligation to buy an asset at a fixed price K at a fixed date T.

To price this contract imagine we are at time t and the current price of an asset is S, we want to buy a forward contract with maturity T and fixed price K.

We have two options, we can buy the contract at time t for a price of F and at time T pay K to get the asset. Alternatively we can buy the asset for S and borrow $e^{-r(T-t)}K$

at time t, so at time T we will pay $K = e^{r(T-t)}e^{-r(T-t)}K$ while having the asset. Both options have the same cashflow at time T, pay K and own the asset, so either we have

$$F = S - e^{-r(T-t)}K,$$

or there is an arbitrage opportunity.

We have found the price of the forward contract by replicating its cashflows, using the underlying asset and the risk-free rate, plus arguing that there should not be an arbitrage opportunity.

3.1.2 Example: Put-Call parity

Definition 3.1 (European call option)**.** An **European call option** is a financial contract that gives the holder the right, but not the obligation, to **buy** an asset at a fixed price (the strike price) at a fixed date (the maturity date).

The payoff of an European call option at maturity T is given by:

$$C_T = \max(S_T - K, 0), \tag{11}$$

where S_T is the price of the underlying asset at maturity and K is the strike price.

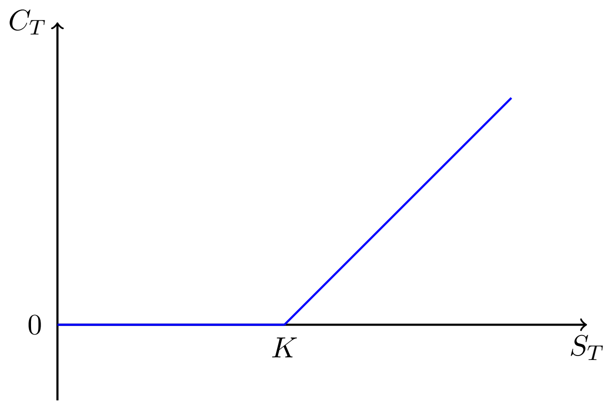

Figure 1: Payoff of a European call option

Definition 3.2 (European put option)**.** An **European put option** is a financial contract that gives the holder the right, but not the obligation, to **sell** an asset at a fixed price (the strike price) at a fixed date (the maturity date).

The payoff of an European put option at maturity T is given by:

$$P_T = \max(K - S_T, 0), \tag{12}$$

where S_T is the price of the underlying asset at maturity and K is the strike price.

Now consider a portfolio where we buy a call option and sell a put option with same strike K and maturity T, the payoff of such portfolio at time T will be exactly $S_T - K$. So we are essentilly reciving the asset and paying K at time T, as we have seen this is the payoff of a forward contract, which means that at time t the price of our portfolio must be equal to the price of a forward of the asset,

$$C_t - P_t = F_t = S_t - e^{-r(T-t)}K, \tag{13}$$

this formula is called *put-call parity* and illustrates the relationship between the price of a call and a put option.

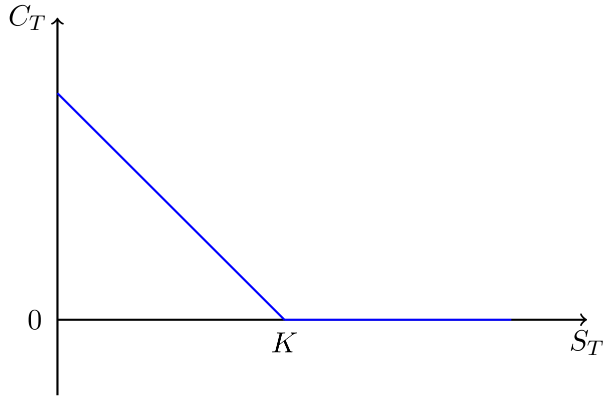

Figure 2: Payoff of a European put option

3.2 The Black-Scholes model

The Black-Scholes model is a mathematical model for pricing options, that is based on replicating the payoffs of the option dynamically using the underlying asset and a risk-free rate bond.

Like before we assume that the market is arbitrage-free, and that there is a constant risk-free rate r. As the model is based on dynamic replication, we also need to assume that we can rebalance our replicating portfolio continuously, i.e. we can buy and sell any amount of the underlying asset and the risk-free rate bond instantaneously at any time.

Let S_t be the price of the underlying asset at time t and β_t be the price of the risk-free rate bond at time t. The dynamics of this two assets are given by the following stochastic differential equations (SDEs):

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \qquad d\beta_t = r\beta_t dt.$$

To construct the replicating portfolio. Let a_t be the amount of the underlying asset in our replicating portfolio and b_t be the amount of the risk-free rate bond in our replicating portfolio. Then the value of the portfolio at any time t is given by:

$$V_t = a_t S_t + b_t \beta_t.$$

We need this portfolio to have the same payoff as the option at time T, so

$$V_T = \max(S_T - K, 0).$$

As the portfolio is dynamically replicating the option, the cashflow of the portfolio must match the ones of the option, these are the payoff as seen before, and the initial payment, so no cash must come in or out of the portfolio at any time between t and T. This means that the portfolio must be *self-financing*, so the rebalanced must come from selling the underlying asset and buying the bond, or vice versa. This is expressed mathematically as

$$dV_t = a_t dS_t + b_t d\beta_t, \tag{14}$$

which can be extended to

$$dV_t = (a_t \mu S_t + b_t r \beta_t)\, dt + a_t \sigma S_t dW_t.$$

Now as the value of the replicating portfolio V_t should be equal to the value of the option at any time t, we assume that this can be expressed by a function $f(t, S_t)$, depending only on the time t and the price of the underlying asset S_t. Thus applying Ito's formula we have that

$$dV_t = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S}\mu S_t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S_t^2\right) dt + \frac{\partial f}{\partial S}\sigma S_t dW_t,$$

matching the two expressions for dV_t we have that

$$a_t = \frac{\partial f}{\partial S},$$

and

$$b_t = \frac{1}{r\beta_t}\left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S_t^2\right)$$

as V_t is the value of the replicating porfolio and the option at time t, we have that

$$f(t, S_t) = V_t = a_t S_t + b_t \beta_t,$$

using the values of a_t and b_t from above,

$$f(t, S_t) = \frac{\partial f}{\partial S}S_t + \frac{1}{r}\left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S_t^2\right),$$

that can be rearranged to

$$\frac{\partial f}{\partial t} = rf(t, S_t) - r\frac{\partial f}{\partial S}S_t - \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S_t^2, \tag{15}$$

with terminal condition

$$V_T = f(T, S_T) = \max(S_T - K, 0),$$

which is the **Black-Scholes PDE**.

3.3 Pricing - Solving the Black-Scholes PDE

There is an analytical solution to the Black-Scholes PDE. Our derivation follows the approach of [9, Ch. 11.4], adapted to log-moneyness coordinates $x = \ln(S/K)$ instead of the log-price $\ln S$ used there; the two differ only by the constant shift $\ln K$ and the subsequent reduction to the heat equation is structurally identical.

Start by change of variables. Let $\tau = T - t$ be the time to maturity, and $x = \ln\left(\frac{S}{K}\right)$ be the log-moneyness. We define a new function $u(\tau, x) = f(t, S)$ so the Black-Scholes PDE is transformed to,

$$\frac{\partial u}{\partial \tau} = \frac{1}{2}\sigma^2\frac{\partial^2 u}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial u}{\partial x} - ru(\tau, x).$$

Now we would like to remove the drift term (the derivative with respect to x) and the and the zero order term, so we have a standard heat equation. To do it we do another transformation,

$$u(x, \tau) = e^{\alpha x + \beta\tau}v(x, \tau),$$

where α and β are the coeficients that lead to the heat equation, using,

$$\begin{aligned}
u_\tau &= e^{\alpha x+\beta\tau}(\beta v + v_\tau),\\
u_x &= e^{\alpha x+\beta\tau}(\alpha v + v_x),\\
u_{xx} &= e^{\alpha x+\beta\tau}(\alpha^2 v + 2\alpha v_x + v_{xx}),
\end{aligned}$$

we get,

$$v_\tau = \frac{1}{2}\sigma^2 v_{xx} + v_x\left(\sigma^2\alpha + r - \frac{1}{2}\sigma^2\right) + v\left(\frac{1}{2}\sigma^2(\alpha^2-\alpha) + r\alpha - r - \beta\right),$$

thus,

$$\alpha = \frac{1}{2} - \frac{r}{\sigma^2} = -\frac{1}{\sigma^2}(r - \frac{1}{2}\sigma^2),$$

and

$$\begin{aligned}
\beta &= \alpha\left(r + \frac{1}{2}\sigma^2(\alpha-1)\right) - r\\
&= \alpha\left(\frac{1}{2}r - \frac{1}{4}\sigma^2\right) - r\\
&= -\frac{1}{2\sigma^2}\left(r - \frac{1}{2}\sigma^2\right)^2 - r\\
&= -\frac{1}{2\sigma^2}\left(r + \frac{1}{2}\sigma^2\right)^2.
\end{aligned}$$

So we get the standard heat equation,

$$\frac{\partial v}{\partial \tau} = \frac{1}{2}\sigma^2\frac{\partial^2 v}{\partial x^2},$$

with initial conditions for v,

$$v(x,0) = e^{-\alpha x}u(x,0) = Ke^{-\alpha x}\max(e^x - 1, 0).$$

Now we can solve the equation using the heat operator,

$$v(x,\tau) = \int_{-\infty}^{\infty} v(y,0)k(\sigma^2;\tau,x-y)dy,$$

where the function k is the *heat kernel*, i.e. the fundamental solution of the heat equation $v_\tau = \lambda v_{xx}$, given by

$$k(\lambda;t,x) := \frac{1}{\sqrt{4\lambda\pi t}}\exp\left(-\frac{x^2}{4\lambda t}\right), \tag{16}$$

see [9, Ch. 11.4]. Using the initial condition we get,

$$v(x,\tau) = \int_0^{\infty} Ke^{-\alpha y}(e^y - 1)k(\frac{\sigma^2}{2};\tau,x-y)dy,$$

this leaves us with two integrals

$$\begin{aligned}
I_1 &= K\int_0^{\infty} e^{(1-\alpha)y}k(\frac{\sigma^2}{2};\tau,x-y)dy,\\
I_2 &= -K\int_0^{\infty} e^{-\alpha y}k(\frac{\sigma^2}{2};\tau,x-y)dy,
\end{aligned}$$

if we make a change of variable, $s = x - y$, then

$$I_1 = Ke^{(1-\alpha)x} \int_{-\infty}^{x} e^{-(1-\alpha)s} k(\frac{\sigma^2}{2}; \tau, s) ds,$$
$$I_2 = -Ke^{-\alpha x} \int_{-\infty}^{x} e^{\alpha s} k(\frac{\sigma^2}{2}; \tau, s) ds.$$

To evaluate both integrals we use the following identity, whose proof is a standard completion of the square.

Lemma 3.1. Let Φ denote the standard Gaussian cumulative distribution function,

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-u^2/2} du.$$

Then for any $a, b \in \mathbb{R}$ and $\sigma, \tau > 0$,

$$\int_{-\infty}^{a} k(\frac{\sigma^2}{2}; \tau, s) e^{-bs} ds = e^{\frac{b^2\sigma^2\tau}{2}} \Phi\left(\frac{a}{\sigma\sqrt{\tau}} + b\sigma\sqrt{\tau}\right). \tag{17}$$

Proof. Completing the square in the exponent of the integrand,

$$-\frac{s^2}{2\sigma^2\tau} - bs = -\frac{(s + b\sigma^2\tau)^2}{2\sigma^2\tau} + \frac{b^2\sigma^2\tau}{2},$$

so pulling out the constant factor and applying the change of variable $u = (s+b\sigma^2\tau)/(\sigma\sqrt{\tau})$,

$$\int_{-\infty}^{a} k(\tfrac{\sigma^2}{2}; \tau, s) e^{-bs} ds = e^{\frac{b^2\sigma^2\tau}{2}} \int_{-\infty}^{a} \frac{1}{\sqrt{2\pi\sigma^2\tau}} \exp\left(-\frac{(s + b\sigma^2\tau)^2}{2\sigma^2\tau}\right) ds$$
$$= e^{\frac{b^2\sigma^2\tau}{2}} \int_{-\infty}^{\frac{a+b\sigma^2\tau}{\sigma\sqrt{\tau}}} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = e^{\frac{b^2\sigma^2\tau}{2}} \Phi\left(\frac{a}{\sigma\sqrt{\tau}} + b\sigma\sqrt{\tau}\right).$$

□

Applying Lemma 3.1 to I_1 (with $a = x$, $b = -(1-\alpha)$) and to I_2 (with $a = x$, $b = \alpha$), and substituting back to $u = e^{\alpha x + \beta\tau} v$, we get

$$I_1 \Rightarrow I_3 = K \exp\left(\alpha x + \beta\tau + (1-\alpha)x + \frac{(1-\alpha)^2\sigma^2\tau}{2}\right) \Phi(d_1),$$
$$I_2 \Rightarrow I_4 = -K \exp\left(\alpha x + \beta\tau - \alpha x + \frac{\alpha^2\sigma^2\tau}{2}\right) \Phi(d_2),$$

where

$$d_1 = \frac{x}{\sigma\sqrt{\tau}} + (1-\alpha)\sigma\sqrt{\tau},$$
$$d_2 = \frac{x}{\sigma\sqrt{\tau}} - \alpha\sigma\sqrt{\tau} = d_1 - \sigma\sqrt{\tau},$$

rearranging and simplifying the expressions

$$I_3 = K \exp\left(x + \left(\beta + \frac{(1-\alpha)^2\sigma^2}{2}\right)\tau\right) \Phi(d_1),$$
$$I_4 = -K \exp\left(\left(\beta + \frac{\alpha^2\sigma^2}{2}\right)\tau\right) \Phi(d_2),$$

the exponent of the first term simplifies to

$$x+\left(\beta+\frac{(1-\alpha)^2\sigma^2}{2}\right)\tau=x+\frac{1}{2\sigma^2}\left(-\left(r+\frac{1}{2}\sigma^2\right)^2+\left(r+\frac{1}{2}\sigma^2\right)^2\right)\tau=x,$$

and for the second one we use the fact that while deriving β, on the second equality we found

$$\beta=\frac{1}{2}\alpha\left(r-\frac{1}{2}\sigma^2\right)-r$$

multiplying by $1=\frac{\sigma^2}{\sigma^2}$ and rearranging we get

$$\beta=\frac{\sigma^2}{2}\alpha\frac{1}{\sigma^2}\left(r-\frac{1}{2}\sigma^2\right)-r$$

and using the definition of α we get

$$\beta=-\frac{\sigma^2}{2}\alpha^2-r.$$

Thus

$$\begin{aligned}I_3&=Ke^{x}\Phi(d_1),\\ I_4&=-Ke^{-r\tau}\Phi(d_2).\end{aligned}$$

Finally, reverting the substitutions $\tau=T-t$ and $x=\ln(S/K)$ we get the **Black-Scholes formula**

$$f(t,S)=S\Phi(d_1)-Ke^{-r(T-t)}\Phi(d_2), \tag{18}$$

where

$$\begin{aligned}d_1&=\frac{\ln\left(\frac{S}{K}\right)}{\sigma\sqrt{T-t}}+(1-\alpha)\sigma\sqrt{T-t}\\ &=\frac{\ln\left(\frac{S}{K}\right)}{\sigma\sqrt{T-t}}+\left(1+\frac{1}{\sigma^2}\left(r-\frac{1}{2}\sigma^2\right)\right)\sigma\sqrt{T-t}\\ &=\frac{\ln\left(\frac{S}{K}\right)}{\sigma\sqrt{T-t}}+\left(\frac{1}{\sigma^2}\left(\sigma^2+r-\frac{1}{2}\sigma^2\right)\right)\sigma\sqrt{T-t}\\ &=\frac{1}{\sigma\sqrt{T-t}}\left(\ln\left(\frac{S}{K}\right)+\left(r+\frac{1}{2}\sigma^2\right)(T-t)\right),\end{aligned}$$

and, of course

$$\begin{aligned}d_2&=d_1-\sigma\sqrt{T-t}\\ &=\frac{1}{\sigma\sqrt{T-t}}\left(\ln\left(\frac{S}{K}\right)+\left(r-\frac{1}{2}\sigma^2\right)(T-t)\right).\end{aligned}$$

3.4 Calibration - Implied volatility

The Black-Scholes formula (18) gives the call price as a function of market observables (t,S,r), contract parameters (K,T), and the single volatility parameter σ. In practice, market prices $C^{\text{mkt}}(K,T)$ are observable for a range of strikes and maturities, and the question is: which value of σ is implied by each observed price?

Definition 3.3 (Implied volatility)**.** Given a market price C^{mkt} for a European call with strike K and maturity T, the **implied volatility** $\hat{\sigma}(K,T)$ is the unique $\sigma>0$ such that

$$C_{\text{BS}}(t,S;\,K,T,\,\hat{\sigma})=C^{\text{mkt}}, \tag{19}$$

where C_{BS} denotes the Black-Scholes formula (18).

Uniqueness follows from strict monotonicity of the call price in σ. To see this we compute the *vega* of the option, $\mathcal{V} := \partial C_{\mathrm{BS}}/\partial\sigma$.

Write $\tau = T - t$ for brevity and note that d_1 and d_2 both depend on σ. Splitting d_1 into its two terms makes differentiation straightforward,

$$d_1 = \frac{\ln(S/K) + r\tau}{\sigma\sqrt{\tau}} + \frac{1}{2}\sigma\sqrt{\tau},$$

so

$$\frac{\partial d_1}{\partial\sigma} = -\frac{\ln(S/K) + r\tau}{\sigma^2\sqrt{\tau}} + \frac{\sqrt{\tau}}{2}. \tag{20}$$

Since $d_2 = d_1 - \sigma\sqrt{\tau}$,

$$\frac{\partial d_2}{\partial\sigma} = \frac{\partial d_1}{\partial\sigma} - \sqrt{\tau} = -\frac{\ln(S/K) + r\tau}{\sigma^2\sqrt{\tau}} - \frac{\sqrt{\tau}}{2}. \tag{21}$$

In particular,

$$\frac{\partial d_1}{\partial\sigma} - \frac{\partial d_2}{\partial\sigma} = \sqrt{\tau}. \tag{22}$$

Differentiating (18) with the chain rule gives

$$\frac{\partial C_{\mathrm{BS}}}{\partial\sigma} = S\,\phi(d_1)\,\frac{\partial d_1}{\partial\sigma} - Ke^{-r\tau}\phi(d_2)\,\frac{\partial d_2}{\partial\sigma},$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$ is the standard Gaussian density. The two terms simplify if we use the identity

$$S\,\phi(d_1) = Ke^{-r\tau}\phi(d_2). \tag{23}$$

To verify it, note that $d_1 - d_2 = \sigma\sqrt{\tau}$ and $d_1 + d_2 = 2(\ln(S/K) + r\tau)/(\sigma\sqrt{\tau})$, so

$$d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = 2\ln(S/K) + 2r\tau.$$

Therefore, since the $1/\sqrt{2\pi}$ factors cancel in the ratio,

$$\frac{\phi(d_2)}{\phi(d_1)} = \frac{e^{-d_2^2/2}}{e^{-d_1^2/2}} = e^{(d_1^2-d_2^2)/2} = \exp\left(\ln\frac{S}{K} + r\tau\right) = \frac{S}{K}e^{r\tau},$$

which is exactly (23). Applying it and using (22),

$$\mathcal{V} := \frac{\partial C_{\mathrm{BS}}}{\partial\sigma} = S\,\phi(d_1)\left(\frac{\partial d_1}{\partial\sigma} - \frac{\partial d_2}{\partial\sigma}\right) = S\sqrt{T-t}\,\phi(d_1) > 0. \tag{24}$$

Since $\mathcal{V} > 0$, the map $\sigma \mapsto C_{\mathrm{BS}}(\sigma)$ is a strictly increasing bijection from $(0,\infty)$ onto the no-arbitrage interval $\big(\max(S - Ke^{-r(T-t)}, 0),\, S\big)$, so the implied volatility exists and is unique for any market price in this range.

Since (19) has no closed-form solution in σ, the implied volatility is computed numerically. The smoothness and strict positivity of $\mathcal{V}$ make Newton's method an efficient choice: starting from an initial guess $\sigma^{(0)} > 0$, one iterates

$$\sigma^{(n+1)} = \sigma^{(n)} - \frac{C_{\mathrm{BS}}(\sigma^{(n)}) - C^{\mathrm{mkt}}}{\mathcal{V}(\sigma^{(n)})}, \tag{25}$$

until $|C_{\mathrm{BS}}(\sigma^{(n)}) - C^{\mathrm{mkt}}| < \varepsilon$.

3.4.1 The implied volatility surface

If Black-Scholes with a single σ perfectly described the market, every option on the same underlying would yield the same implied volatility. In practice, the map $(K, T) \mapsto \hat{\sigma}(K, T)$ is non-trivial, and the resulting object is called the **implied volatility surface**.

Two features of this surface are well documented in the literature, see [3]:

- **Volatility skew**: for a fixed maturity T, equity index options exhibit a pronounced *negative skew*: implied volatility increases as the strike decreases relative to the spot. This reflects the demand for downside protection and the asymmetry of return distributions.

- **Term structure**: for a fixed strike, implied volatility varies with maturity, reflecting the market's expectation of future realized volatility at different horizons.

The non-flatness of the surface is direct empirical evidence that constant-σ Black-Scholes is inconsistent with observed market prices. This motivates more flexible models such as the *local volatility model*, which replaces σ by a deterministic function $\sigma(t, S)$ calibrated to reproduce the full surface exactly, as described in the following section.

Figure 3 shows the implied volatility surface computed from SPY option quotes on 2026-04-30, using the Black-Scholes-Merton formula with a continuously compounded dividend yield calibrated per maturity from put-call parity (see Appendix A for implementation details). The characteristic features are clearly visible: the volatility skew in the moneyness direction and the term structure in the maturity direction. Figure 4 shows the smile slice at the median maturity ($T \approx 0.21$ yr), where OTM puts (red) and OTM calls (blue) meet smoothly at the money.

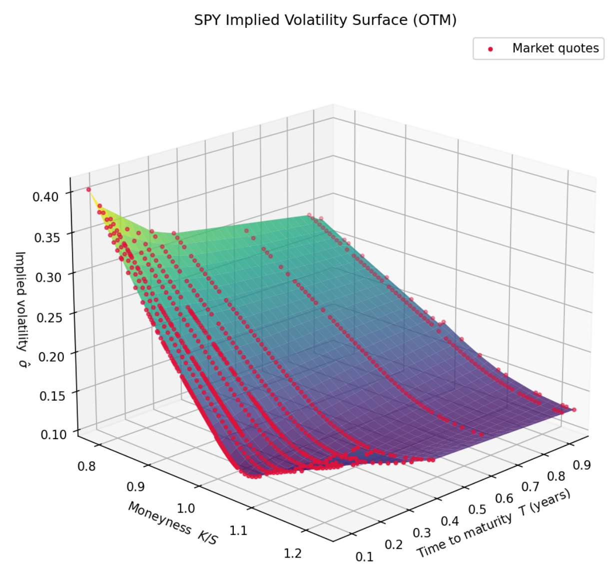

Figure 3: SPY implied volatility surface (OTM composite) on 2026-04-30. Axes: time to maturity T (years), moneyness K/S (inverted scale), implied volatility $\hat{\sigma}$. Red dots are individual market quotes; the surface is linearly interpolated.

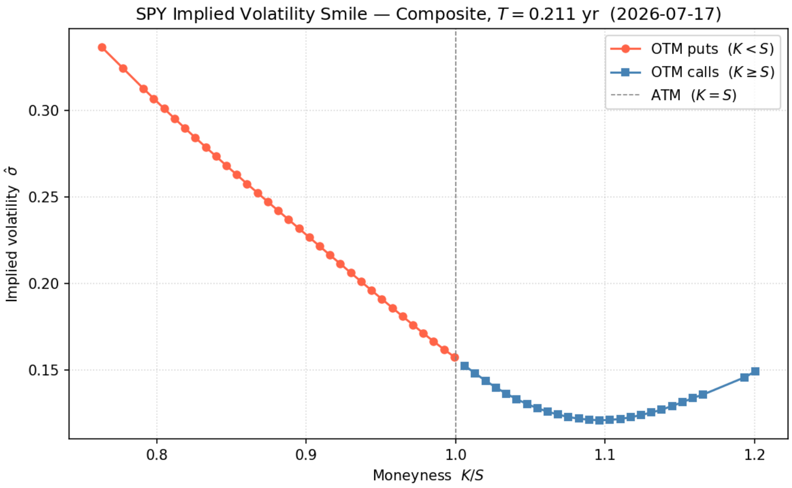

Figure 4: Volatility smile at the median maturity ($T \approx 0.21$ yr, expiration 2026-07-17). OTM puts (red) and OTM calls (blue) are joined from the calibrated composite surface. The smooth join at $K = S$ reflects the consistent dividend yield calibration.

4 Arbitrage theory

We can generalize the approach used to derive the Black-Scholes PDE for a more general products Thus in this section will try to give a general framework to derive equations to price a *contingent claim*.

Definition 4.1 (Market)**.** A market is a set of $n+1$ assets $S_0, S_1, \ldots, S_n$, where the price of S_0 is given by

$$\frac{dS_0(t)}{S_0} = r(S_0(t), t)dt,$$

and the prices of the other assets are given by

$$dS_i(t) = \mu(S_i(t), t)dt + \sigma_i(S_i(t), t)^\intercal dW(t),$$

where $W(t)$ is a k-dimensional Brownian motion.

Definition 4.2 (Portfolio)**.** A portfolio in the market $\{S_i(t)\}$ is an $n+1$ dimensional $\mathcal{F}_t^{(k)}$-adapted stochastic process

$$\theta(t) = (\theta_0(t), \theta_1(t), \ldots, \theta_n(t)),$$

an its value at time t is given by

$$V(t) = \sum_{i=0}^{n} \theta_i(t) S_i(t).$$

Note that as a portfolio $\theta(t)$ is an adapted process it doesn't use future information, so it is a generalization on what in reality would be to rebalance ones investments in the market at time t.

We can generalize the self-financing condition presented in (14) as,

$$V(t) = V(0) + \int_0^t \theta(u) dS(u), \tag{26}$$

in differential form

$$dV(t) = \theta(u) dS(t). \tag{27}$$

A contingent claim is a financial instrument whose payoff depends on the outcome of uncertain future events, for example an European option. As our market consist of assets that are descried as stochastic processes, an European contingent claim paying at time T is a function $f(S(T), T)$ depending on these assets, which is a random variable as $S(T)$ is a random vector of the values of the assets at time T.

4.1 PDE method

Consider an European contingent claim with payoff $f(S(T))$, similar to what we did in section 3.2, we want to construct a replicating portfolio $V(S(t), t)$ that has the same payoff as the contingent claim at time T. Thus using the Ito formula (5) we have

$$\begin{aligned} dV(S(t), t) &= \frac{\partial V}{\partial t} dt + \sum_{i=0}^{n} \frac{\partial V}{\partial S_i} dS_i(t) + \frac{1}{2} \sum_{i,j} \frac{\partial V^2}{\partial S_i \partial S_j} dS_i(t) dS_j(t) \\ &= \sum_{i=0}^{n} \frac{\partial V}{\partial S_i} dS_i(t) + \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sum_{i,j} \frac{\partial V^2}{\partial S_i \partial S_j} \sigma_i(S(t), t)^\intercal \sigma_j(S(t), t) \right) dt, \end{aligned}$$

imposing the self-financing condition (27) we find that

$$\theta_i(t) = \frac{\partial V}{\partial S_i}, \quad \forall i = 0, \dots, n, \tag{28}$$

and

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sum_{i,j} \frac{\partial V^2}{\partial S_i \partial S_j} \sigma_i(S(t), t)^\intercal \sigma_j(S(t), t) = 0. \tag{29}$$

Also using $V(S(t), t) = \theta(t)S(t)$ in (28) we get

$$V(S(t), t) = \sum_{i=0}^{n} \frac{\partial V}{\partial S_i} S_i. \tag{30}$$

The equations (29) and (30) describe the PDE for the price of the contingent claim, using as boundary condition the payoff at time T

$$V(S(T), T) = f(S(T)).$$

As in Black-Scholes model, μ does not explicitly appear in the PDE, its only present implicitly in the prices $S(t)$ of the assets.

4.2 Martingale method

The martingale method is an alternative framework to the PDE method, that allows to price contingent claims in a more flexible way.

We will use chapter 10 of [2] as a reference for this section, we start with a couple of definitions related with the concept of market and portfolio defined previously.

Definition 4.3 (Attainable process)**.** We say a price process $V(t)$ is *attainable* if there exists a self-financing portfolio $\theta(t)$ such that $V(t) = \theta(t)S(t)$.

The attainable process is a formalization of the dynamic replicating portfolio idea. An other restriction on the portfolios we are going to consider is the following,

Definition 4.4 (Admissible portfolio)**.** A portfolio $\theta(t)$ is called *admissible* if exists $\alpha > 0$ such that

$$\int_0^t \theta(t) dS(t) \geq -\alpha, \quad \forall t \in [0, T].$$

In real life we can interpret this condition as a limit on the amount the portfolio can lose or borrow, as bank, clearing house or counterparty will not allow such risks and would force to cancel the deal. Technically also allow to cancel out some other undesirable strategies that would prevent some key theorems to hold, see [9] sections 14.5 and 14.6 for more details on such strategies.

Now we will formalize the idea of arbitrage in a market.

Definition 4.5 (Arbitrage)**.** A self-financing and admissible portfolio $\theta(t)$ is an *arbitrage* if one of the following conditions hold for time t:

1. $\theta(0)S(0) < 0$ and $P(\theta(t)S(t) \geq 0) = 1$,
2. $\theta(0)S(0) = 0$ and $P(\theta(t)S(t) \geq 0) = 1$ and $P(\theta(t)S(t) > 0) > 0$,

Consider that the *risk-free* asset S_0 has a constant price of 1, i.e. the *risk-free* rate is $r(S(t), t) \equiv 0$. Under this assumption we introduce the *risk-neutral measure* $\mathcal{Q}$ as,

Definition 4.6. A probability measure $\mathcal{Q}$ on $\mathcal{F}_T$ is called a *risk-neutral measure* (also known as *equivalent martingale measure*) if

- $\mathcal{Q}$ is equivalent to P on $\mathcal{F}_T$.
- All price process in the market $S_0, S_1, \ldots, S_n$ are martingales under $\mathcal{Q}$ on the time interval $[0, T]$.

We refer to P as the "real" probability measure and reflects the true probability of the price processes, whereas $\mathcal{Q}$ is a probability measure under which the expected price of any asset is equal to the current one regardless of their risk (volatility).

Under this assumption we have the following theorem.

Theorem 4.1 (First fundamental theorem of asset pricing)**.** The market $\{S_i(t)\}$ is arbitrage-free if and only if there exists an equivalent (local) martingale measure $\mathcal{Q}$.

We include the local martingales as restricting to the martingales would eliminate too many processes, that are of interest.

The restriction on the risk-free asset being constant can be dropped by considering the first price process a numeraire and normalizing the market. For S_0 to be a numeraire we need that $S_0(t) > 0$ almost surely for all $t \geq 0$. Using S_0 as numeraire we can consider the processes $Z_i = S_i(t)/S_0(t)$, which are the *discounted* price processes of the assets.

Definition 4.7 (Normalized market)**.** A normalized market is a vector of processes Z given by

$$Z_i(t) = \frac{S_i(t)}{S_0(t)}, \quad i = 1, \ldots, n.$$

The results we have presented so far are invariant to the usage of the "regular" market $\{S_i(t)\}$ as defined in (4.1) or the normalized market $\{Z_i(t)\}$ (4.7). So if a portfolio is self-financing on $\{S_i(t)\}$ then it is also self-financing on $\{Z_i(t)\}$ it is also self-financing on $\{Z_i(t)\}$, and if a portfolio does not have arbitrage on $\{S_i(t)\}$ then it is also self-financing on $\{Z_i(t)\}$ it does not have it on $\{Z_i(t)\}$ either. Thus 4.1 holds for any market $\{S_i(t)\}$ by just normalizing it.

Notice that if the numeraire is the risk-free asset S_0 then the the market $\{S_i(t)\}$ is arbitrage-free if all assets $S_i(t)$ have a local rate of return $r(t)$ under $\mathcal{Q}$, that is, if $\mu_i(S_i(t), t) = r(t)S_i(t)$ for all $i = 1, \ldots, n$ under $\mathcal{Q}$.

Under the assumption of a market with no arbitrage, a key question is wether a contingent claim can be replicated by a portfolio consisting on the market assets. This yields the following theorem.

Theorem 4.2 (Second fundamental theorem of asset pricing)**.** Assuming absence of arbitrage, the market model is complete if and only if the martingale measure $\mathcal{Q}$ is unique.

Finally we can give a general formula to price contingent claims. Under the assumption of a complete market, the price of a contingent claim $f(S(T))$ is given by

$$V(t, f(S(T))) = \mathbb{E}^{\mathcal{Q}} S_0(t) \left[\frac{f(S(T))}{S_0(T)} | \mathcal{F}_t \right] \tag{31}$$

in case that the numeraire is the risk-free asset,

$$V(t, f(S(T))) = \mathbb{E}^{\mathcal{Q}} \left[e^{\int_t^T r(s)ds} f(S(T)) | \mathcal{F}_t \right]$$

5 Local Volatility Model

We are going to introduce the Local Volatility Model mainly following [1].

5.1 Model description

The **local volatility model** is the simplest and most widely used volatility model in derivatives pricing. It extends the Black-Scholes framework by allowing the instantaneous volatility to be a deterministic function of both time and the spot price of the underlying asset,

$$\sigma = \sigma(t, S).$$

So the dynamics of the underlying asset under the local volatility model are given by the following stochastic differential equation,

$$\frac{dS_t}{S_t} = \mu(t, S_t)dt + \sigma(t, S_t)dW_t,$$

and the pricing PDE is identical to the Black-Scholes PDE, with the deterministic volatility function replacing the constant volatility

$$\frac{\partial f}{\partial t} = rf(t, S_t) - r\frac{\partial f}{\partial S}S_t - \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2(t, S_t)S_t^2,$$

5.1.1 Market model

A market model is a valuation model that includes vanilla options as part of the underlying, this means we are assuming that when hedging a derivative, we can use the vanilla options as a hedging instrument, in other words, the derivative can be replicated using a combination of the underlying as well as vanilla options. Thus vanilla options under a market model are exactly priced.

The local volatility model is a market model, since it is calibrated to match the vanilla option prices observed in the market

5.2 Dupire Equation

We will deduce the Dupire equation from arbitrage free arguments.

5.3 Calibration problem

In both the Black-Scholes model and the Local Volatility model, the parameter must be calibrated to market data to make the model usable. We will focus on the calibration of the Volatility function $\sigma(S, t)$ in the Local Volatility model, and will assume that the risk-free rate r and the dividend yield q are known and constant.

The market data consists of option prices for different strikes K and maturities T, So the calibration problem is to find a function $\sigma(S, t)$ such that the model prices $C(S_0, t_0, T, K; \sigma)$ match the observed in the market. While the volatility function could be any function, usually it is assumed to be smooth and have some regularity.

5.4 Calibration of the Local Volatility Model - Inverse PDE problem

We will present a calibration method based on the inverse problem following [6]

5.4.1 Change of variable - Log-moneyness

Market option prices are quoted over a discrete set of strikes K and maturities T, but the absolute level of the strike depends on the current price of the underlying asset S_0. To avoid this dependency and make all options comparable, we will work with what is called the *log-moneyness*, defined as

$$y = \log(K/S_0),$$

the *log-moneyness* indicates how far in or out of the money is an option in logarithmic scale.

We redefine the notation of the volatility function and the call option price to reassemble a more standard for PDEs, let us define $\tau = T$ the time to maturity, $a(y, \tau) = \frac{1}{2}\sigma^2(K, T)$ as the diffusion parameter, and $u(\tau, y) = C(S_0, t_0; T, K)$ the call option price, to do this change of variable to the Dupire equation first we compute the derivatives of C in terms of u, with respect to T,

$$C_T = \frac{\partial u}{\partial \tau} = u_\tau,$$

with respect to K we have to apply the chain rule,

$$C_K = \frac{\partial u}{\partial y}\frac{\partial y}{\partial K} = u_y \frac{1}{K},$$

and,

$$C_{KK} = \frac{\partial}{\partial K}\frac{u_y}{K} = \frac{1}{K}\frac{\partial u_y}{\partial K} - \frac{u_y}{K^2} = \frac{1}{K^2}(u_{yy} - u_y),$$

thus the Dupire PDE becomes,

$$u_\tau = a(y, \tau)(u_{yy} - u_y) - (r - q)u_y + qu. \tag{32}$$

We consider the PDE on a bounded domain $\Omega = (\alpha, \beta) \times (0, T]$, with Dirichlet boundary conditions,

$$u(\tau, \alpha) = \Psi_1(\tau), \quad u(\tau, \beta) = \Psi_2(\tau), \tag{33}$$

and initial condition,

$$u(0, y) = f(y) = S_0 \max(1 - e^y, 0). \tag{34}$$

Thus the inverse problem is to identify the diffusion parameter $a(y, \tau)$ from the initial value boundary problem defined by (32), (33), (34).

5.4.2 Reconstruction algorithm

The proposed algorithm on [6] considers to solve the problem on periods of maturities $[T_i, T_{i+1}]$, for each maturity with option price data.

The idea is to consider the to consider the option price function $f(y)$ at the beginning of the period T_i and $g(y)$ at the end of the period T_{i+1}, then we consider a linear operator A_f that maps the diffusion parameter a to the price function at the end of the period,

$$A_f(a) = u_a(T_{i+1}, y) = g(y),$$

and we want to find a such that this equality holds.

This approach transforms the nonlinear inverse problem into a series of linear problems, that are solvable using a Newton-type method.

5.4.3 Linear operator - Gateaux derivative

To apply Newton-like methods we first need to compute the Gâteaux derivative of the operator A_f. This will give a linear approximation of the operator around a given point, that we will use to transform our nonlinear problem into a series of linear problems.

Proposition 5.1. Let $a \in L^\infty(\Omega)$ be a coefficient such that the forward problem admits a unique solution u_a. The Gâteaux derivative of A_f at a in the direction h, denoted by $A'_f(a)$, is given by $A'_f(a)h = w(\cdot, T)$, where $w(y,\tau)$ is the unique solution to the linearized variational problem:

$$-w_\tau + a(y,\tau)(w_{yy} - w_y) - (r-q)w_y + qw = -h(y,\tau)((u_a)_{yy} - (u_a)_y), \tag{35}$$

subject to homogeneous initial and boundary conditions:

$$w(0,y) = 0, \qquad y \in (\alpha, \beta), \tag{36}$$

$$w(\tau,\alpha) = 0, \quad w(\tau,\beta) = 0, \qquad \tau \in [0,T]. \tag{37}$$

5.4.4 Adjoint operator

To implement gradient-based minimization algorithms efficiently, we introduce the adjoint of the Gâteaux derivative, referred as the backpropagation operator in [6].

Let $A'_f(a)^* : L^2(\Omega) \to L^2(\Omega \times [0,T])$ denote the adjoint operator. For a given residual $\psi \in L^2(\Omega)$ (e.g., $\psi = A_f(a) - g$), the action of the adjoint operator is defined via the solution to a backward parabolic problem.

Proposition 5.2 (Adjoint Operator)**.** Let u_a be the solution to the forward problem for a given volatility parameter a. For any $\psi \in L^2(\Omega)$, let $m(y,\tau)$ be the solution to the following adjoint backward parabolic equation:

$$\frac{\partial m}{\partial \tau} + \frac{\partial^2 a(y,\tau)m}{\partial y^2} + \frac{\partial a(t,\tau)m}{\partial y} + (r-q)\frac{\partial m}{\partial y} = 0, \tag{38}$$

subject to the final and boundary conditions:

$$m(y,T) = \psi(y), \qquad y \in (\alpha, \beta), \tag{39}$$

$$m(\alpha,\tau) = 0, \quad m(\beta,\tau) = 0, \qquad \tau \in [0,T). \tag{40}$$

Then, the adjoint operator applied to ψ is given by:

$$A'_f(a)^*\psi = m\left(\frac{\partial^2 u_a}{\partial y^2} - \frac{\partial u_a}{\partial y}\right). \tag{41}$$

In the case where the volatility parameter $a(y)$ is assumed to be time-independent, the adjoint operator maps into $L^2(\alpha,\beta)$ by integrating the time-dependent sensitivity over the duration of the option:

Corollary 5.3 (Time-Independent Case)**.** If $a = a(y)$, the gradient of the objective function with respect to a is given by:

$$A'_f(a)^*\psi = \int_0^T m(\cdot,\tau)\left(\frac{\partial^2 u_a}{\partial y^2}(\cdot,\tau) - \frac{\partial u_a}{\partial y}(\cdot,\tau)\right) d\tau. \tag{42}$$

5.4.5 Algorithm

As mentioned before the time dimension is not used in this procedure, instead what we do is discretize the time from t to T into intervals and forward propagate the function u using A_f for each interval, then we compute the residual with respect to g at T and use the backpropagation operator to compute the gradients for each interval, backwards, and use them to update the parameter a. The calibration algorithm proceeds as follow:

Set the functional spaces for a and u.
Set the boundary and initial conditions for the forward problem.
Initialize the volatility parameter a.
Compute the target option price function $g(y)$ from market data.
while $\|u_a(y) - g(y)\| > \textit{tolerance}$ **do**
  $u_a \leftarrow A_f(a)$
  $\nabla_a^* \leftarrow A_f'(a)^*(u_a - g)$
  $a \leftarrow$ Update a using gradient descent step.
end while

5.4.6 Practial implementation

We have implemented the calibration algorithm in *Python* using the *FENICS* library, so we use finite elements to solve the forward and backward PDEs.

Also we used *scipy* to implement the optimization algorithm, in particular we used the *L-BFGS-B* algorithm to perform the minimization of the objective function.

First we needed to transform the PDEs into their variational forms, we multiply the equation (32) by a test function $v \in H_0^1(\Omega)$ and integrate over Ω,

$$\int_\alpha^\beta u_\tau v dy = \int_\alpha^\beta a u_{yy} v dy - \int_\alpha^\beta (a + r - q) u_y dy + \int_\alpha^\beta quvdy.$$

Using that $(au_y)_y = au_{yy} + a_y u_y$ we can rewrite the second order as,

$$-\int_\alpha^\beta a u_{yy} v dy = \int_\alpha^\beta a_y u_y v dy, -\int_\alpha^\beta (au_y)_y v dy$$

then integrating by parts,

$$-\int_\alpha^\beta (au_y)_y v dy = \int_\alpha^\beta a u_y v_y dy - [a u_y v]_\alpha^\beta,$$

since $v \in H_0^1(\Omega)$ the boundary terms vanish, $[au_y v]_\alpha^\beta = 0$. Moreover for practical implementation we would like to avoid the derivative with respect to τ, as we will use discrete time increments, so we can rewrite the left hand side as,

$$\int_\alpha^\beta u_\tau v dy = \int_\alpha^\beta \frac{\partial u}{\partial \tau} v dy + \int_\alpha^\beta u \frac{\partial v}{\partial \tau} dy = \frac{d}{d\tau} \int_\alpha^\beta uvdy,$$

as v does not depend on τ, $\frac{\partial v}{\partial \tau} = 0$. Thus the variational form of the forward equation is given by,

$$\frac{d}{d\tau} \int_\alpha^\beta uvdy = \int_\alpha^\beta a u_y v_y dy + \int_\alpha^\beta (a + r - q) u_y v dy - \int_\alpha^\beta quvdy. \tag{43}$$

For equation (38), we proceed as before, multiplying by a test function $v \in H_0^1(\Omega)$ and integrating over Ω,

$$\int_\alpha^\beta m_\tau v dy = \int_\alpha^\beta (am)_{yy} v dy + \int_\alpha^\beta (am)_y v dy + \int_\alpha^\beta (r-q) m_y v dy,$$

then we integrate by parts the second order term,

$$\int_\alpha^\beta (am)_{yy} v dy = -\int_\alpha^\beta (am)_y v_y dy + \left[(am)_y v\right]_\alpha^\beta,$$

the boundary term vanishes since $v \in H_0^1(\Omega)$. And for the first order terms we have,

$$\int_\alpha^\beta (am)_y v dy = -\int_\alpha^\beta am v_y dy + \left[amv\right]_\alpha^\beta,$$

as before the boundary term vanishes. We do the same treatment for the last term,

$$\int_\alpha^\beta (r-q) m_y v dy = -\int_\alpha^\beta (r-q) m v_y dy.$$

So, also using the same treatment for m_τ as we did before, the variational form of the adjoint equation is given by,

$$\frac{d}{d\tau}\int_\alpha^\beta m v dy = -\int_\alpha^\beta (am)_y v_y dy - \int_\alpha^\beta am v_y dy - \int_\alpha^\beta (r-q) m v_y dy. \tag{44}$$

5.4.7 Numerical results

5.4.8 Alternative approach Interpolation/Exaptrapolation of the IV surface - Optional

A Implied Volatility Surface - Implementation Details

This appendix describes the pipeline used to construct the SPY implied volatility surface shown in Section 3.2, including the practical difficulties encountered and the assumptions made at each step. All code lives in Implied Volatility Code and is reproducible from the cached market snapshot dated 2026-04-30.

A.1 Data

SPY option chains are fetched via the `yfinance` library. We restrict to maturities $4\,\text{wk} \leq T \leq 1\,\text{yr}$ and require a minimum trading volume of 10 contracts per quote. When both bid and ask are positive the mid-price is used; otherwise the last traded price is taken as a fallback (relevant for pre-market snapshots where the spread is stale but the last price is more informative). Data for calls and puts are cached separately as *parquet* files so that all subsequent computations are fully reproducible without network access.

A.2 Risk-free rate

The risk-free rate is taken from the US Treasury par yield curve published daily by the US Department of the Treasury. Par yields are available at standard maturities (1M, 2M, 3M, 4M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, 30Y) and are converted to continuously-compounded rates via

$$r_{\text{cc}} = \ln\left(1 + \frac{y}{100}\right),$$

where y is the par yield in percent. Rates are linearly interpolated (in T) between the published nodes and held flat outside the range. Figure 5 shows the resulting curve on the reference date.

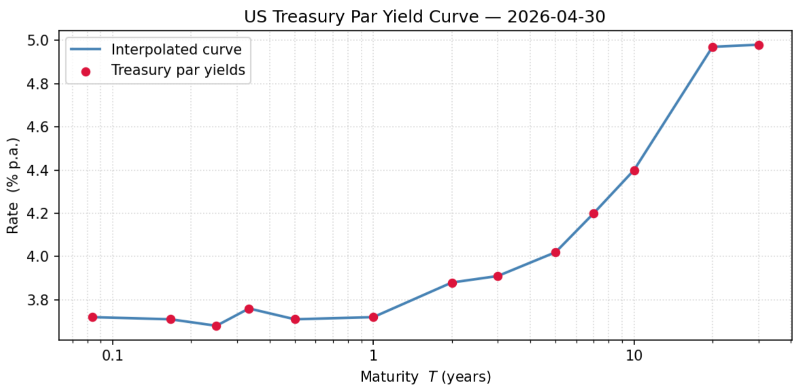

Figure 5: US Treasury par yield curve on 2026-04-30 (log-maturity axis). Red dots are published nodes; the line is the linear interpolant used to assign a per-maturity discount rate.

Note that the risk-free rate could be refined further as the US Treasuries are not strictly risk-free, as they carry a small credit risk premium. A better discounting rate for SPY options would be the overnight index swap (OIS) rate rather than the Treasury curve, but for our purposes the difference is not enough to justify the additional complexity of fetching and processing OIS data.

A.3 Dividend yield calibration

SPY pays dividends quarterly, so any pricing formula must account for the dividend yield q. Getting q wrong breaks the internal consistency of the surface: puts and calls at the same strike and maturity imply different volatilities, producing a visible discontinuity at the money in any composite surface. We went through three successive approximations before reaching a satisfactory result.

Stage 1 - $q = 0$. The simplest assumption is to ignore dividends entirely. This is equivalent to treating SPY as a non-dividend-paying asset, which is a standard simplification when the dividend yield is small relative to the interest rate and the maturity is short. For maturities beyond a few months, however, cumulative dividends are non-negligible (SPY's trailing yield is approximately 1.1%) and the forward price implied by put-call parity,

$$F = Se^{(r-q)T},$$

is overestimated if q is set to zero. As a result, the Black-Scholes-Merton formula assigns different implied volatilities to calls and puts at the same strike, since each type is sensitive to the forward in a different direction.

Stage 2 - Fixed annual yield. The natural correction is to use the trailing twelve-month dividend yield reported by the data provider ($q \approx 1.14\%$ for SPY on the reference date). This constant is applied uniformly to all maturities. The surface improves near the money, but the fix is imprecise: SPY dividends are paid on a quarterly schedule, so the effective yield between two option expirations depends on how many ex-dividend dates fall within that window. A short-dated option expiring before the next ex-dividend date carries an effective $q \approx 0$, while a longer-dated option spanning several dividend payments accumulates a yield closer to the annual figure. Applying a single annual constant systematically over-corrects for short maturities and under-corrects in an uncontrolled way across the term structure.

Stage 3 - Put-Call parity calibration. The cleanest approach is to infer q at each expiration directly from market prices using put-call parity, keeping the Treasury rate r fixed. For each matched call-put pair at strike K_i and expiration T, put-call parity gives

$$C_i - P_i = Se^{-qT} - K_i e^{-rT}.$$

Defining the cost-of-carry forward $F = Se^{(r-q)T}$ and multiplying both sides by e^{rT} yields an implied forward per pair,

$$F_i = (C_i - P_i)\, e^{rT} + K_i,$$

where q is encoded inside F_i rather than appearing explicitly. Taking the median over the relevant pairs at a given expiration yields a robust forward estimate $\tilde{F}$, and inverting $\tilde{F} = Se^{(r-q)T}$ gives

$$q = r - \frac{\ln(\tilde{F}/S)}{T}.$$

A key implementation choice is *which pairs to include.* Put-call parity holds model-free for any European options, but the quality of the forward estimate depends on how cleanly each leg is priced. For a strike $K < S$ the call is in-the-money and the put is out-of-the-money; for $K > S$ the roles are reversed. Deep in-the-money options are significantly less liquid: their price is dominated by intrinsic value, the bid-ask spread is wide relative to the time value, and last-trade prices can be hours stale. Feeding these pairs into the forward estimate biases $\tilde{F}$ toward noisy values.

We therefore restrict the calibration to near-ATM pairs satisfying $|K/S - 1| \leq 0.05$ (i.e. strikes within $\pm 5\%$ of the current spot). In this band, both the call and the put are close to at-the-money; both legs have tight spreads and meaningful open interest, and the time value is large relative to the bid-ask noise. The median over these pairs (rather than the mean or a least-squares fit) guards against the occasional stale quote even in the near-ATM region. If fewer than two near-ATM pairs are available the constraint is relaxed to all matched pairs.

Since r is fixed from the Treasury curve, there is only one free parameter per expiration, which avoids the instability of simultaneously fitting both r and q. Calibrated values are bounded to $q \in [-0.02, 0.10]$; if the result is outside this range (a sign of too few matched pairs or erroneous data), the provider value is used as a fallback.

The resulting term structure shows $q \approx 0$ for near-term expirations (the next quarterly dividend has not yet been declared within the window) and a gradual convergence to the trailing annual yield for longer maturities, which is economically consistent with the quarterly payment schedule.

Figure 6 illustrates the three stages for the December 2026 expiration ($T \approx 0.63$ yr), where the effect of the dividend yield is large enough to be clearly visible. Each panel overlays the put-implied and call-implied volatility curves across the full strike range. With $q = 0$ (Stage 1) the two curves diverge significantly and the level is shifted upward for OTM puts and downward for OTM calls. With a fixed annual q (Stage 2) the ATM region improves but the term-structure mismatch remains. With the PCP-calibrated q (Stage 3) the two curves sit on top of each other near the money, confirming that the composite surface can be joined smoothly. The residual divergence at deep in-the-money strikes visible in all three panels is due to liquidity issues (wide bid-ask spreads, stale prices) rather than the dividend yield assumption, and is removed by restricting the composite to out-of-the-money quotes only.

A.4 OTM composite construction

Once r and q are consistent with market prices, the composite surface is formed by taking out-of-the-money options only: OTM puts ($K < S$) for the left wing and OTM calls ($K \geq S$) for the right wing.

The rationale is that in-the-money options are significantly less liquid than their out-of-the-money counterparts: ITM bids tend to be stale, spreads are wide, and the option price is dominated by intrinsic value, so a small absolute pricing error translates into a large relative error in the time value — and therefore a large error in the inverted implied volatility. This is visible in Figure 6: the divergence between call and put IV curves away from the money persists even in Stage 3, not because q is wrong but because deep-ITM prices are unreliable.

Figures 7 and 8 show the full call and put surfaces (all strikes, no ITM filter) to illustrate the noise. The clean composite (Figure 3) is obtained after restricting to $0.75 \leq K/S \leq 1.25$ and keeping only the OTM leg for each option type.

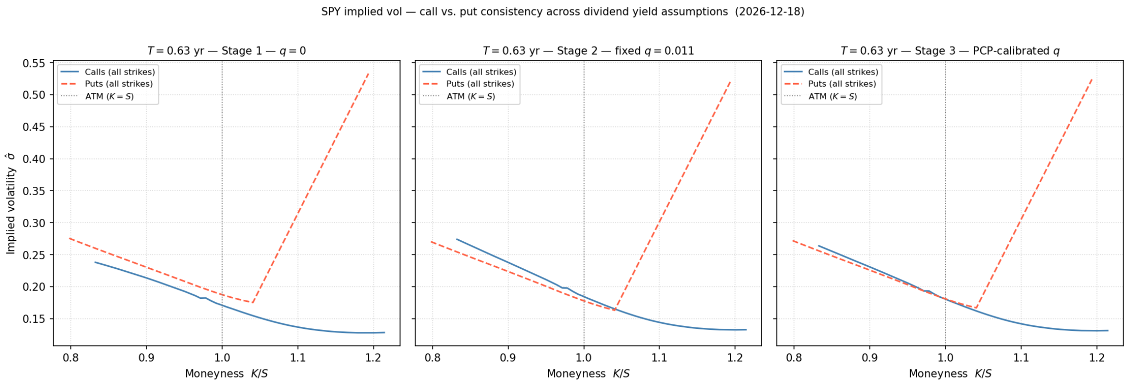

Figure 6: Call-implied (solid blue) and put-implied (dashed red) volatility across all available strikes at the December 2026 expiration ($T \approx 0.63$ yr), under three dividend yield assumptions. Stage 1 ($q = 0$): curves diverge sharply — the forward is overestimated, shifting call and put IVs in opposite directions. Stage 2 (fixed annual q): improved near ATM, but the flat term structure leaves systematic errors for maturities where the true q differs from the annual average. Stage 3 (PCP-calibrated q from near-ATM pairs): the two curves coincide in the OTM region, where both legs are liquid. The residual divergence far from ATM (deep ITM options) is a liquidity artefact — those strikes are excluded from the composite surface.

A.5 Expiration selection

SPY has a dense cluster of weekly expirations in the short end (roughly 1-8 weeks). Including all available expirations would over-weight this region and produce a surface that appears smooth in absolute maturity but poorly samples the longer end. To obtain a balanced representation, the full maturity range is divided into $N = 10$ log-spaced bins; within each bin the expiration with the highest combined open interest (calls plus puts) is kept. This ensures that liquidity guides the selection and that the resulting sample is approximately uniform in $\log T$, which is the natural scale on which the term structure of volatility varies.

A.6 Summary of assumptions

For reference, the key modelling assumptions are:

- **European exercise**: SPY options are technically American-style. The early-exercise premium is small for an index ETF with a low continuous yield and is neglected here; the BSM formula is used as an approximation.
- **Continuous dividends**: actual SPY dividends are discrete quarterly payments. The PCP calibration implicitly converts them into a continuous yield that is consistent with the current option prices, which is the standard market convention.
- **Moneyness filter** $0.75 \leq K/S \leq 1.25$: strikes outside this range are too illiquid to contribute reliable IV observations.
- **Volume filter**: quotes with fewer than 10 contracts of daily volume are excluded.
- **Single snapshot**: the surface is computed from a single end-of-day snapshot; intraday bid-ask movements are not captured.

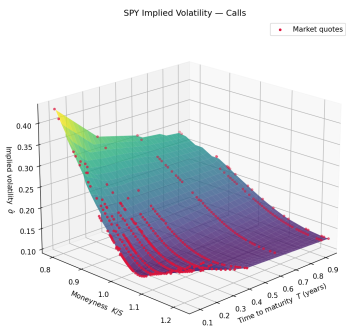

Figure 7: IV surface from all call quotes (including deep ITM, $K \ll S$). Deep ITM calls carry noisy implied volatilities due to wide spreads and stale prices.

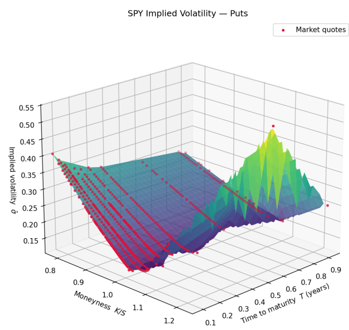

Figure 8: IV surface from all put quotes (including deep ITM, $K \gg S$). Same liquidity issue for strikes far above the spot.

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